

Where was MH370 at 00:19 UTC?

A reproduction of the DSTG Bayesian aircraft-position pdf (Davey et al. 2016, Fig. 10.3)

Hypotheses enabled: radar-fix (parameters in run.json). Sensitivity run 'next-routes', varying altitude_range_ft (not declared) to [25000.0, 43000.0]; bfo_vertical_rate (not declared) to True; early (not declared) to {'cas_envelope_kt': [210.0, 330.0], 'mach_range': [0.45, ...]; mach_range (not declared) to [0.73, 0.84]; fuel (not declared) to {'ceiling': True, 'factor_mean': 1.0004, 'factor_sd': 0.0...; fuel_model (not declared) to 'internal-v1.json'; fuel_tables (not declared) to 'fuel-tables.json'; particles_per_mode [1000000, 500000, 2500000, 500000, 2500000] to [500000, 250000, 1250000, 250000, 1250000]; sampler (not declared) to {'temper_epochs': ['m1839', 'm1941', 'm2041', 'm2141', 'm...; hypotheses radar-fix.

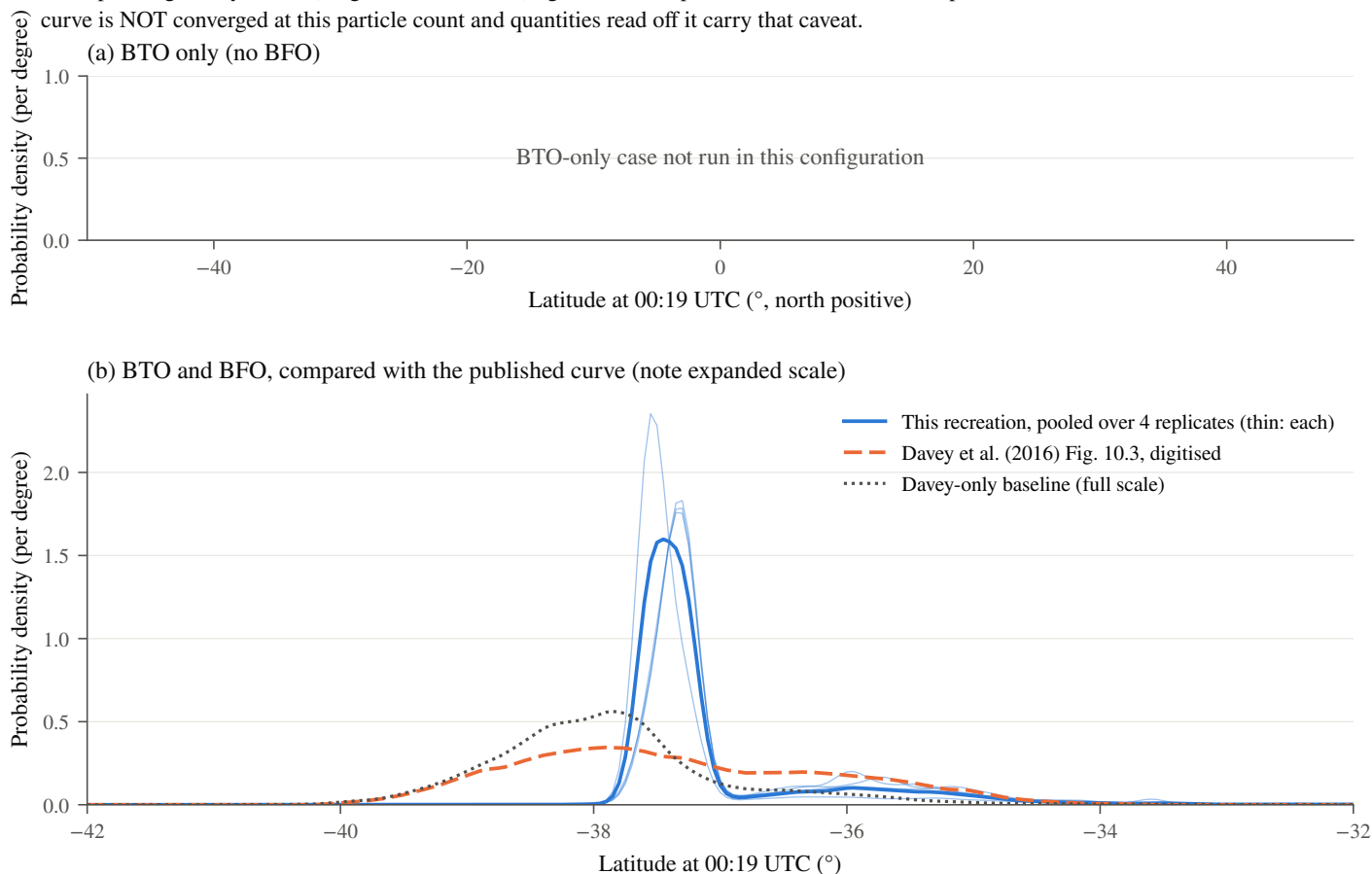
This report recreates the probability density function (pdf) of the aircraft's latitude at the final satellite handshake, 00:19 UTC on 8 March 2014, from the model published by Davey, Gordon, Holland, Rutten and Williams, Bayesian Methods in the Search for MH370 (Springer, 2016). The aircraft state is filtered from the penultimate radar point at 18:01:49 UTC through all Inmarsat burst timing (BTO) and burst frequency (BFO) measurements, using the published cruise, manoeuvre and measurement models.

With BTO and BFO the recreated pdf has median 37.36°S and mode 37.45°S, with 95% of probability between 33.48°S and 37.73°S. The published curve has median 37.53°S, mode 37.90°S and 95% range 34.90–39.38°S.

The two densities overlap by 40%. This curve is centred 0.2° further north and carries 14% of probability in the published northern shoulder between 34.5°S and 36.5°S (25% in the book).

Relative to the Davey-only baseline, the median moves 0.60° north, the 34.5–36.5°S shoulder holds 14% of probability (base 8%), and the two curves overlap by 38%.

The curve pools 4 independent replicates of 3,500,000 particles. Single replicates overlap each other by 63%–96% and their medians span 0.21°. Over all balanced partitions of the 4 replicates the split-half overlap averages only 81.2% (range 79.8% to 82.3%) against the 4-replicate floor of 0.896, so the pooled curve is NOT converged at this particle count and quantities read off it carry that caveat.



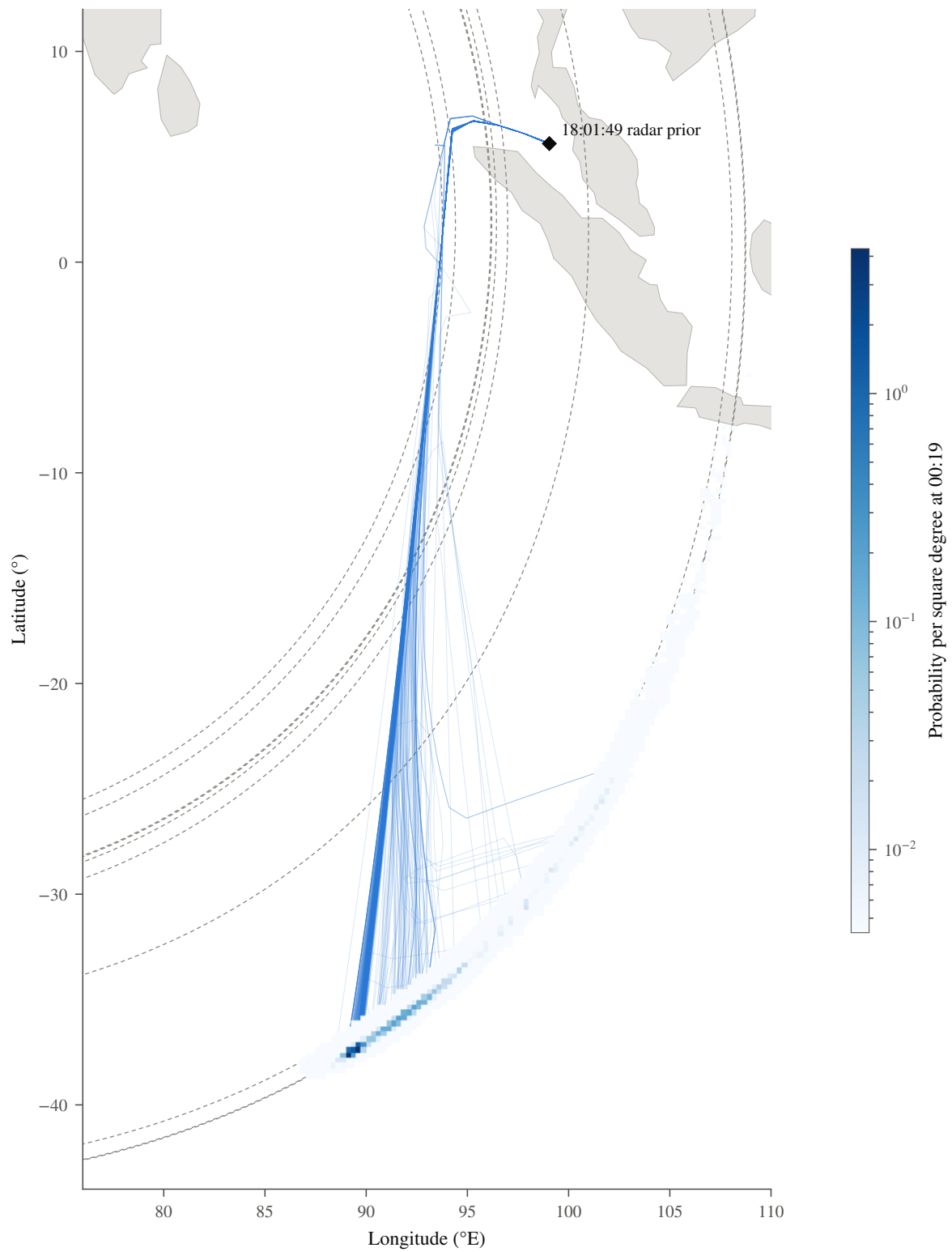
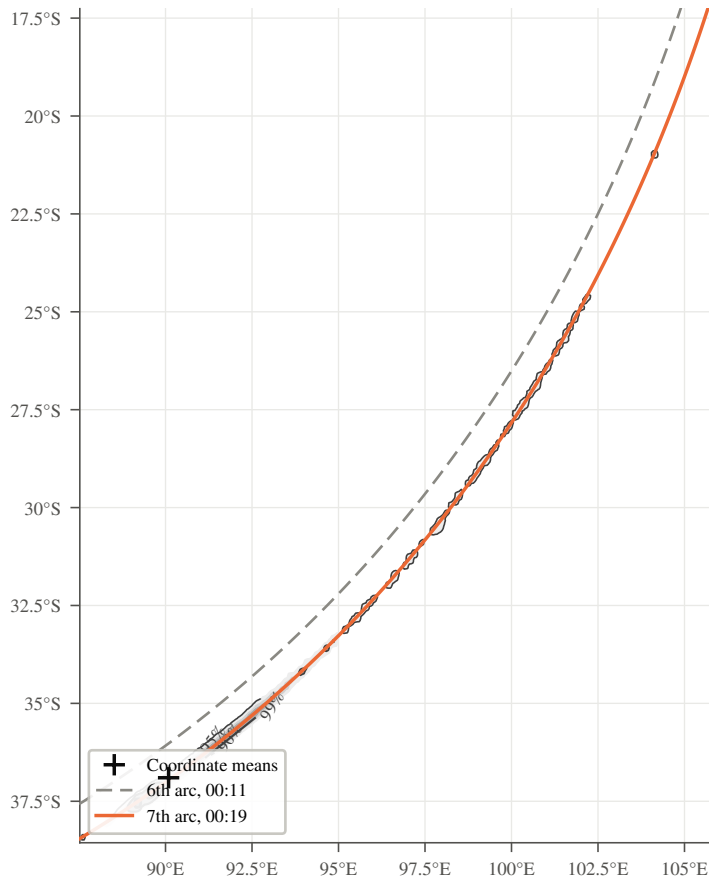


Figure 2. Posterior aircraft position at 00:19 UTC with BTO and BFO, pooled over 4 replicates (blue shading, logarithmic), and 400 routes drawn in proportion to posterior weight from replicate seed 1 (thin lines, 10-minute vertices). Dashed grey curves are the BTO arcs at a reference altitude of 35,000 ft; they are drawn for orientation only. Land: Natural Earth 1:110m.

a Position at 00:19 UTC, over the 99% region



DIFFERENCES FROM DAVEY

Manoeuvre integration step: 5

book: 1

Position sd: 0.5

book: 0.4

Initial Mach set point: -

book: [0.82, 0.03]

Prior mean position and track:

book: not tabulated

BTO noise sd, R600: 63 at 1 epoch

book: 62

Vertical rate in the BFO model:

True

book: not modelled

Weather source: era5-wind-temperature.bin

book: ACCESS-G, 150-500 hPa
(about ...)

Magnetic declination source: igrf14-declination.bin

book: NOAA

Resampling scheme: one filter per autopilot ...

book: randomised branching:
weight ...

Cost-index speed mode:

book: implemented but unused in mos...

Fuel and endurance:

{'factor_mean': 1.0004, '...

book: infinite fuel; constraints
ap...

Fuel proposal: endurance (deadline
unix ...)

book: n/a (no fuel model)

POSTERIOR AT 00:19

median 37.36°S, mode 37.45°S

95% 33.48–37.73°S

replicates 4, split-half 0.740

largest single draw 0.00% of
weight

99% region 14,260 NM²

95% region 5,114 NM²

90% region 2,935 NM²

50% region 231 NM²

b The same posterior, measured from the arc

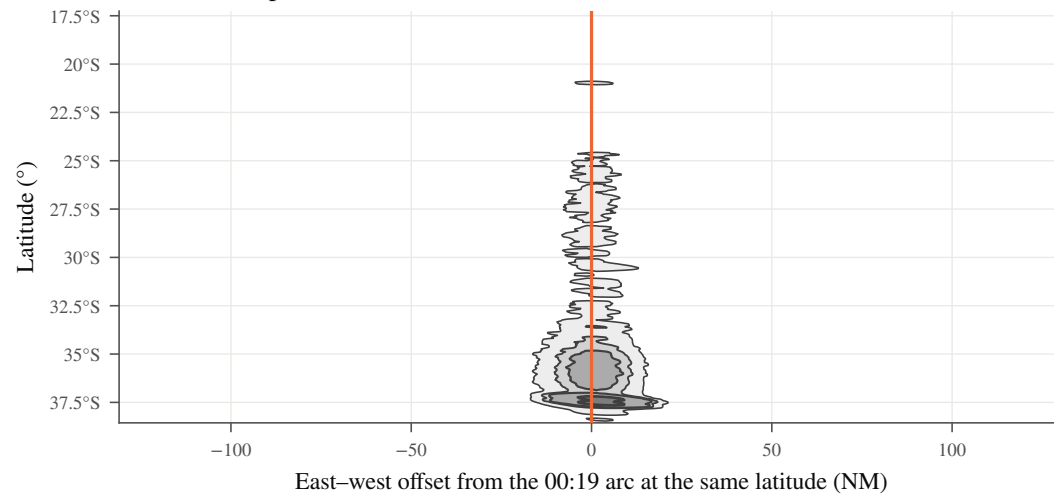
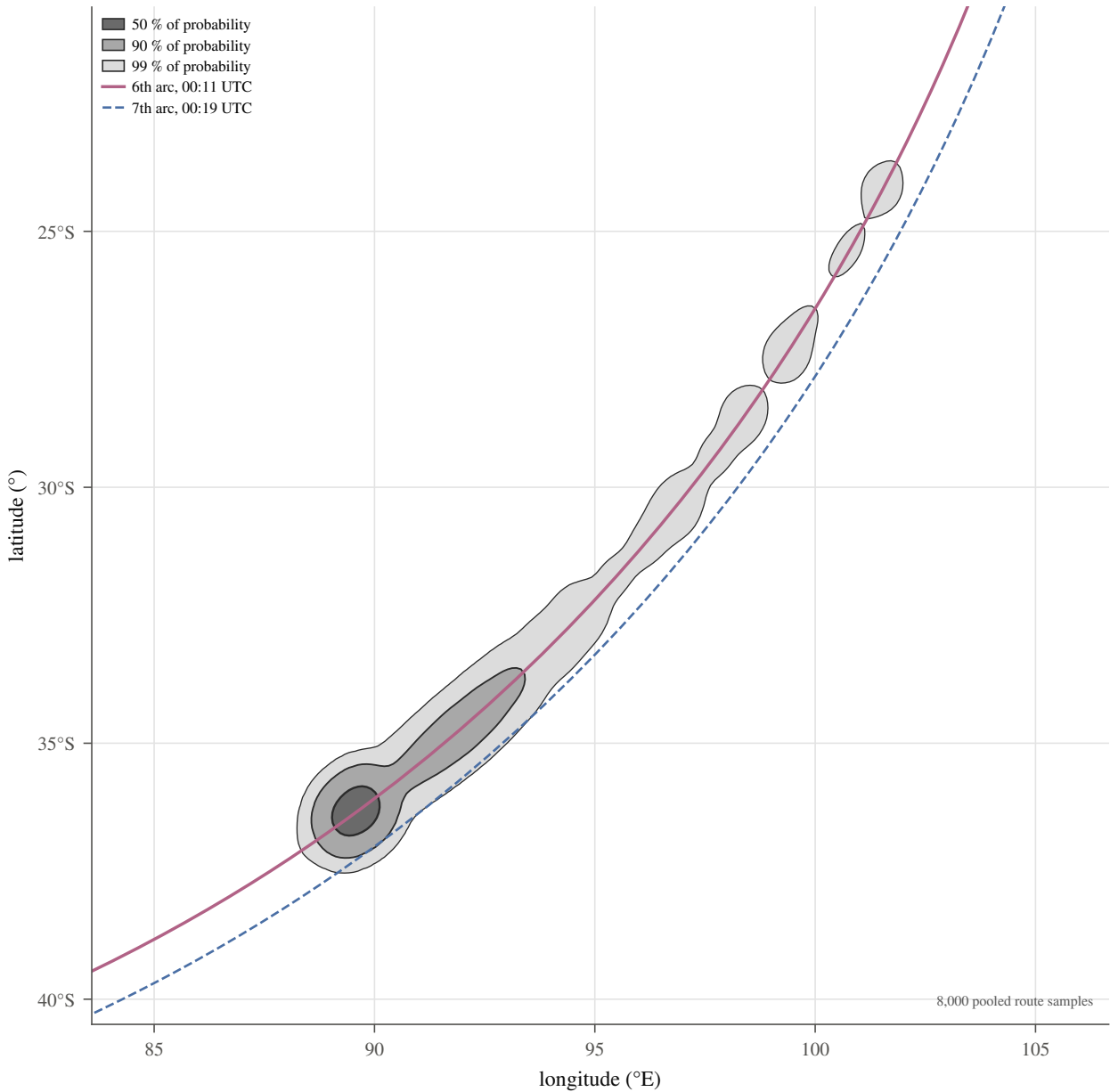


Figure 3. Joint latitude-longitude posterior at 00:19 UTC, drawn as highest-posterior-density regions: the 50% curve bounds the smallest area holding 50% of the probability, and so on outward, so a region here is the smallest place to look for that confidence rather than an interval in either coordinate alone. Binned at 0.05° and smoothed over three bins. Panel a spans the bounding box of the 99% region, so sparse outlying cells are excluded; enclosed areas (NM², at the view's mean latitude) are 99% 14,260, 95% 5,114, 90% 2,935, 50% 231. Panel b replots the same particles against their east-west offset from the 00:19 arc, because the region is about 0.3° wide and 21° long and the inner contours are not separable at a true aspect ratio. Arcs are the reference BTO loci at 35,000 ft; 00:19 is the 7th. Land: Natural Earth 1:110m.

Position at 00:11 UTC

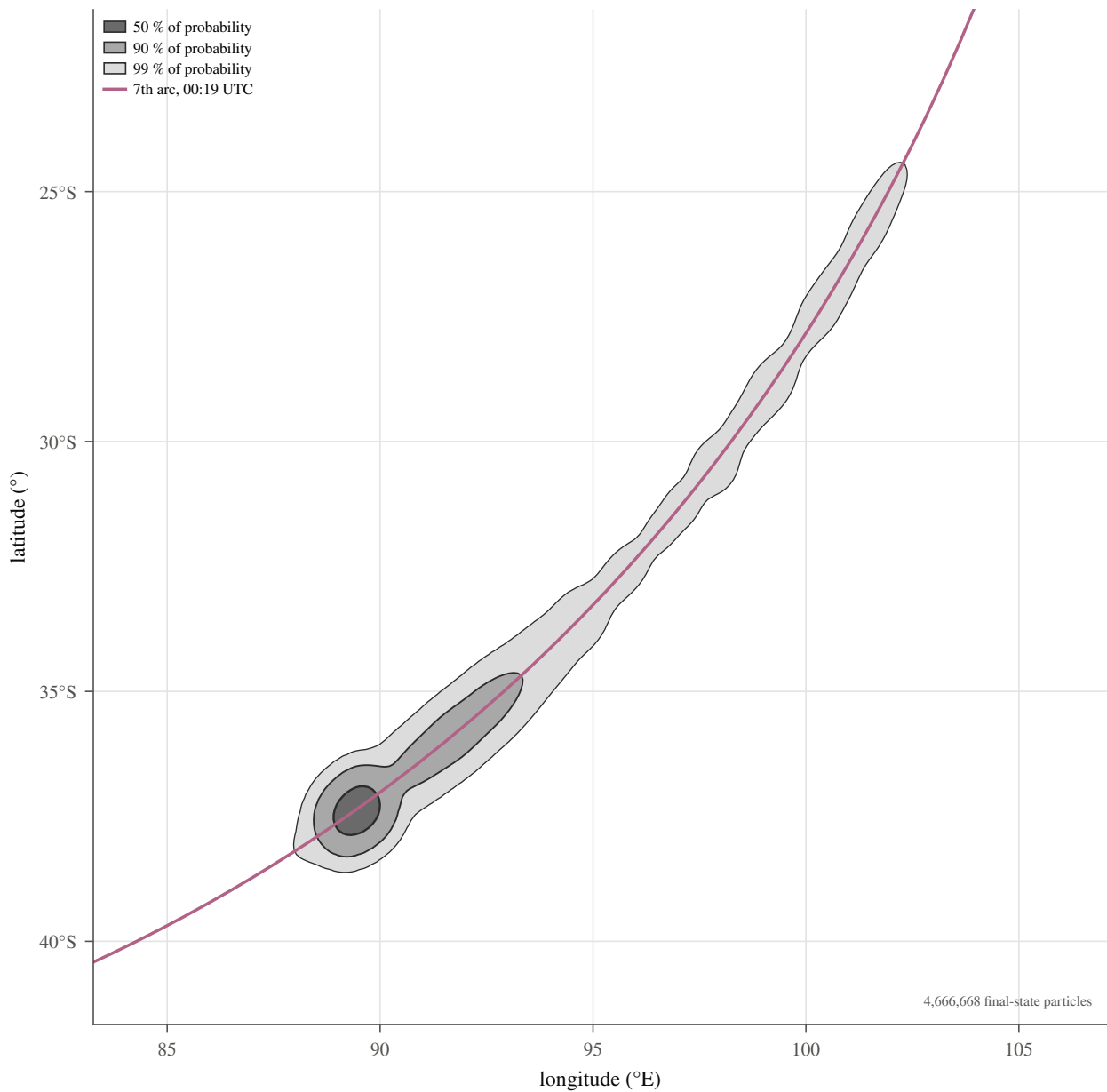
next-routes



next-run-b-routes: Run next-run (b), stratum routes. Stack: davey2016-inmarsat + no-exhaustion-prior + fuel fixes s1-s6 (internal-v1 with temperature term, kappa N(1.0004, 0.0196), 43,800 - kappa x 7,228 kg, weight-dependent ceiling, hard reject, two tanks) + reference-snapshots + radar-full + family overlay; two tanks as bookkeeping only (no one-engine dynamics). One-engine live-engine flow from internal-v1 grid_inop AS DELIVERED, 2x its source tables (core finding 10 Oct): one-engine phases are about half their true length. Baseline overlay: Davey-only baseline. Split-half not converged. PROVISIONAL-OVERNIGHT. Binary 45650e2, x86_64-linux (deskstar), seeds 1, 2, 3, 4, 3,500,000 particles per seed. Prior 18:01:49 (latitude_deg 5.624829, longitude_deg 99.048157, track_deg 289.7, track_sd_deg 1.0). Report built 10 Oct 2026.

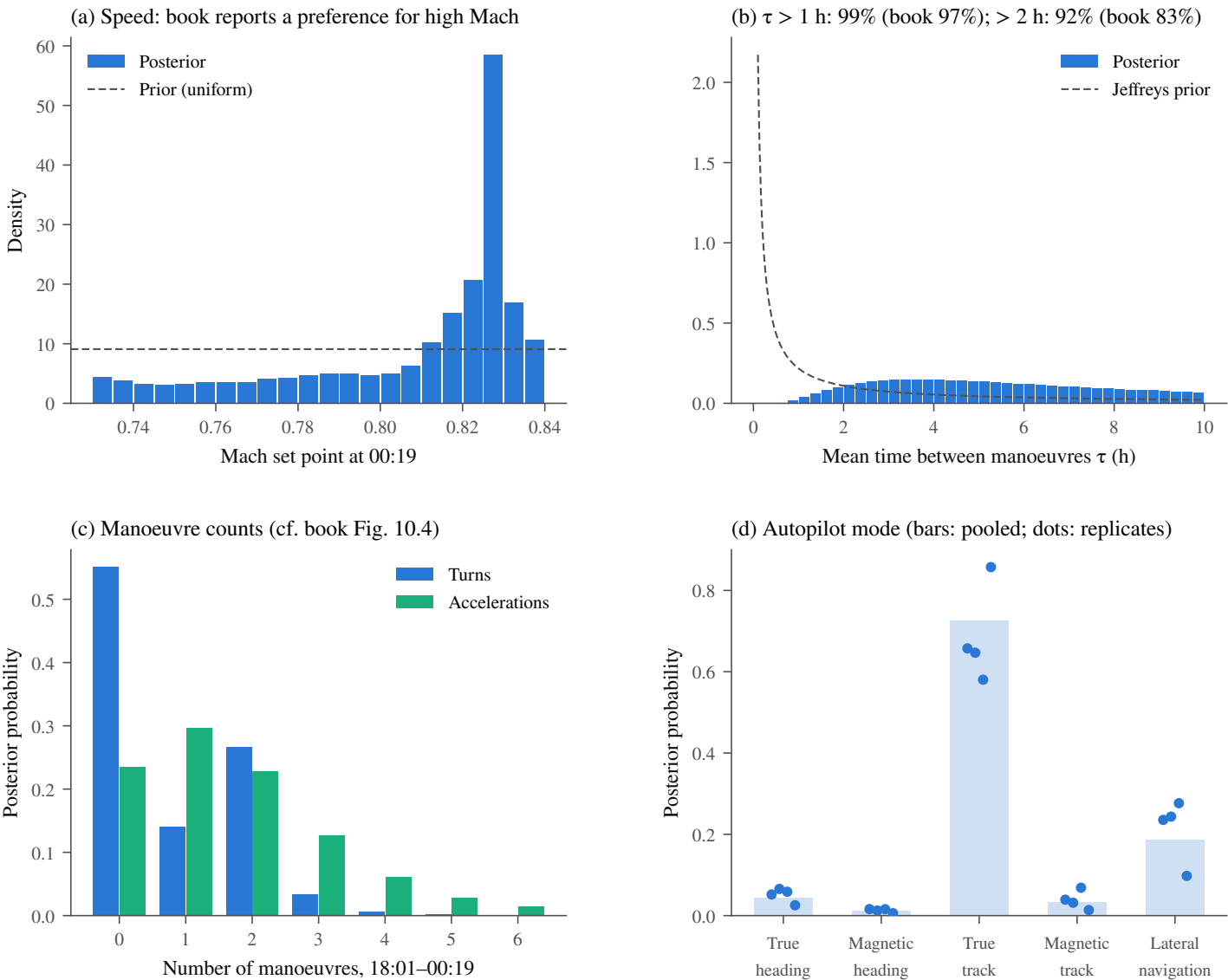
Position at 00:19 UTC

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Posterior behaviour compared with the book



	This recreation	Davey et al. (2016)
Median latitude	37.36°S	37.53°S
Mode	37.45°S	37.90°S
95% interval	33.48–37.73°S	34.90–39.38°S
95% width	4.25°	4.47°
Overlap with published pdf	40% (replicates 33%–46%)	—
Median Mach at 00:19	0.822	upper part of 0.73–0.84
P(north of equator), BTO only	not run	northern mode present

Figure 3 and Table 1. Posterior summaries with BTO and BFO, pooled over replicates, alongside the statements in Davey et al. (2016), sec. 10.2–10.4. Book values of the latitude pdf come from the digitised Fig. 10.3 (bottom); other book values are quoted from the text.

Method, convergence and reproduction

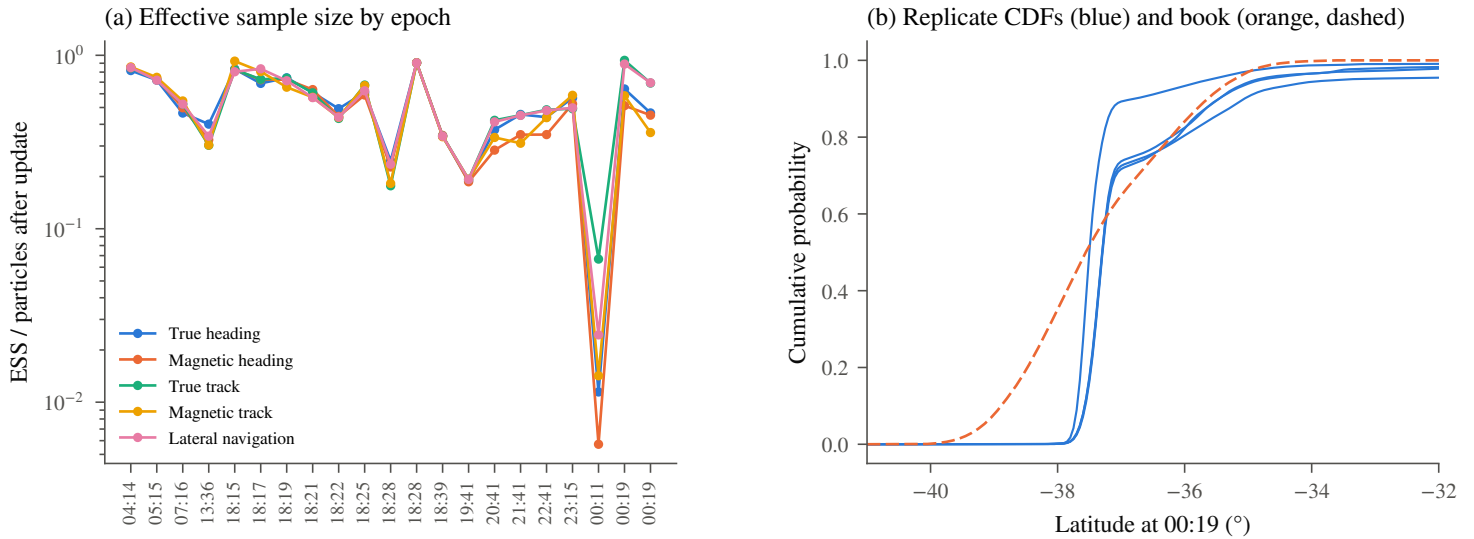


Figure 4. (a) Effective sample size after each measurement update, averaged over replicates, per autopilot-mode filter. The narrowest bottlenecks are the 18:39 C-channel BFO (the southward turn) and the 19:41 BTO arc. (b) Latitude CDFs of the 4 independent replicates; the largest difference between any two is 0.370 (Kolmogorov distance).

Model. The state, dynamics and likelihood follow Davey et al. (2016) chapters 4–8: prior at 18:01:49 UTC with 0.5 NM position and 1° track uncertainty, Mach uniform on 0.73–0.84, altitude on 1,000 ft levels from 25,000 to 43,000 ft; five autopilot modes (true/magnetic heading, true/magnetic track, lateral navigation with a one-off reversion to heading hold); Ornstein–Uhlenbeck fluctuations of Mach, control angle and wind error with the published rates; turns, speed changes and climbs arriving with exponential gaps of common mean τ (Jeffreys prior on 0.1–10 h) at 15° bank, 0.1 Mach per minute and 4,000 ft per minute. BTO is Gaussian with the tabulated per-message errors (29, 43 or 63 μ s); BFO has 7 Hz noise and an unknown constant bias (prior 150 ± 25 Hz) marginalised exactly per particle with a Kalman filter.

Substitutions. The book's ACCESS-G weather is replaced by ERA5 winds and temperatures; magnetic declination is IGRF-14. The prior mean position and track are reconstructed from the book's radar figures because they are not tabulated. Davey et al. used variable-rate branching; this recreation uses fixed-size particle filters with systematic resampling when ESS falls below 50% of the particles. The book warns that conventional resampling collapses static parameters (sec. 10.4). Both are therefore handled exactly: one filter per autopilot mode, combined in proportion to its estimated marginal likelihood, and a Gibbs refresh of τ from its path conditional after every resampling step. Without them, a plain bootstrap filter of one million particles retained only about one hundred distinct prior draws, and an earlier recreation in this project found tails and turn counts that changed with the random seed.

Run. Configuration next-routes; particles per mode 500,000, 250,000, 1,250,000, 250,000, 1,250,000 (order as in Figure 3d), 3,500,000 per replicate; BTO+BFO seeds 1, 2, 3, 4; BTO-only seeds ; code revision 45650e2; 22 threads; total runtime 59.8 min; peak memory 11,802 MiB. Hypotheses enabled: radar-fix (parameters in run.json). Sensitivity run 'next-routes', varying altitude_range_ft (not declared) to [25000.0, 43000.0]; bfo_vertical_rate (not declared) to True; early (not declared) to {'cas_envelope_kt': [210.0, 330.0], 'mach_range': [0.45, ...]; mach_range (not declared) to [0.73, 0.84]; fuel (not declared) to {'ceiling': True, 'factor_mean': 1.0004, 'factor_sd': 0.0...; fuel_model (not declared) to 'internal-v1.json'; fuel_tables (not declared) to 'fuel-tables.json'; particles_per_mode [1000000, 500000, 2500000, 500000, 2500000] to [500000, 250000, 1250000, 250000, 1250000]; sampler (not declared) to {'temper_epochs': ['m1839', 'm1941', 'm2041', 'm2141', 'm...; hypotheses radar-fix.

Reproduce with one command from the repository root: make hypothesis H=radar-fix

Parameters and model choices against Davey et al. (2016)

43 parameters and choices. 31 match the published model; 12 differ and are marked. A difference is not by itself a defect: some are deliberate and documented (ERA5 for the unavailable ACCESS-G, IGRF-14 for NOAA, the sampler), some are the subject of the run (a widened Mach range), and some are worth fixing. Page numbers refer to the book; the full table with citations and notes is written beside this report as parameters.csv.

Parameter	This run	Davey et al. (2016)
Dynamics		
Mach OU reversion rate (1/s)	0.01058	0.01058
Mach OU noise strength (1/s)	2.05e-07	2.05e-07
Control-angle OU reversion rate (1/s)	0.009792	0.009792
Control-angle OU noise strength (rad^2/s)	4.074e-08	4.074e-08
Wind-error OU reversion rate (1/s)	0.001087	0.001087
Wind-error OU noise strength (kn^2/s)	0.07021	0.07021
Manoeuvre time constant, Jeffreys support (h)	[0.1, 10]	[0.1, 10]
Mach after an acceleration, uniform on (Mach)	[0.73, 0.84]	[0.73, 0.84]
Altitude after a vertical manoeuvre, uniform on (ft)	[25000, 43000]	[25000, 43000]
Altitude step (ft)	1000	1000
Bank angle in turns (deg)	15	15
Mach rate during an acceleration (Mach/s)	0.00166667	0.00166667
Vertical rate during a climb or descent (ft/s)	66.6667	66.6667
Mean time to lateral-navigation reversion (s)	31162.2	31162.2
Cruise integration step (s)	10	10
▶ Manoeuvre integration step (s)	5	1
Prior		
Prior epoch (UTC)	2014-03-07T18:01:49Z	2014-03-07T18:01:49Z
▶ Position standard deviation (NM)	0.5	0.4
Track standard deviation (deg)	1	1
BFO bias prior mean (Hz)	150	150
BFO bias prior standard deviation (Hz)	25	25
▶ Initial Mach set point	-	[0.82, 0.03]
▶ Prior mean position and track	-	not tabulated
Autopilot-mode prior weights	[[true heading, 1], [magnetic heading,...	equal over the five modes
Measurement		
BTO fixed offset (us)	495679	495679
BTO noise sd, R1200 (us)	29 at 7 epochs	29
BTO noise sd, anomalous R1200 (us)	43 at 2 epochs	43
▶ BTO noise sd, R600 (us)	63 at 1 epoch	62
BFO noise sd (Hz)	7 at 11 epochs	7
BFO bias treatment	constant per trajectory, Kalman-margina...	constant per trajectory, Kalman-marginalised
Prior		
BFO bias drift rate (Hz^2/s)	0	0
Measurement		
BTO bias treatment	-	fixed per-flight T_channel only; no bias st...
▶ Vertical rate in the BFO model	True	not modelled
00:19 BFO	-	not used; believed unreliable
Model choice		
▶ Weather source	config/../data/era5-wind-temperature.bin	ACCESS-G, 150-500 hPa (about 18,000-44,000...
▶ Magnetic declination source	config/../data/igrf14-declination.bin	NOAA
▶ Resampling scheme	one filter per autopilot mode combined...	randomised branching: weight >= eta branche...
Manoeuvre-type dependence	-	turns, accelerations and vertical manoeuvre...
Turn extent (deg)	-	uniform on -180 to +180
▶ Cost-index speed mode	-	implemented but unused in most experiments;...
▶ Fuel and endurance	{'factor_mean': 1.0004, 'factor_sd': 0....	infinite fuel; constraints applied to the p...
Sampler		
▶ Fuel proposal	endurance (deadline unix 1394237459, pr...	n/a (no fuel model)
Model choice		
Observation cutoff	m0019b	00:19 (BTO only at 00:19)

► marks a difference. Rows whose value is prose or a file identifier cannot be compared by string match; for those the reference file asserts the relationship explicitly and parameters.csv records why. Quantities held per epoch rather than per run — the BTO and BFO measurement standard deviations — are in data/satcom-observations.csv and are compared in the note column of the CSV.

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